

Alex Xiao Cai

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<https://www.linkedin.com/in/axiaocai/> | <https://github.com/alexiao94> | <https://alexiao94.github.io/>

SUMMARY

Full-stack engineer with a background in both **software and mechanical engineering**, specializing in **Django** and **Next.js** systems and **microservices architecture**. Experienced in deploying production workloads on **Azure** and **AWS**, with a track record of reducing latency, deployment time, and operational overhead. Currently seeking onsite, hybrid, or remote roles in the **Austin, TX** area.

SKILLS

Languages: Python, Java, C++, C, C#, Go, JS, TS

Frontend: React, Next.js, Tailwind, Daisy UI

Backend: Django, Spring Boot, FastAPI, ASP.NET, Node.js

Databases: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch

DevOps / Cloud: Docker, Kubernetes, GitHub Actions, Azure, AWS, MinIO

ML / AI: PyTorch, Ollama

Practices: REST APIs, CI/CD, Agile/Scrum, Git, Linux, TDD

Spoken: English, Spanish (Bilingual)

EXPERIENCE

Advanta Seeds

June 2024 - Aug 2025

Software Developer

Remote

- Built responsive web applications with **Next.js** and **Daisy UI**, achieving a **30% improvement in user interaction** through **A/B testing** and close collaboration with UX/UI teams to iterate on design and accessibility.
- Designed and implemented **token-based authentication** with **third-party integrations**, strengthening security across all API endpoints and streamlining the user authentication flow, validated through manual penetration testing and access log review.
- Developed and maintained **Django REST API** endpoints, reducing data latency by **40%** through query optimization and caching strategies, verified through **Postman** benchmarking, **Azure** monitoring tools, and API response time logging before and after implementation, facilitating seamless interaction between modules and external services.
- Implemented and maintained **CI/CD pipelines** using **GitHub Actions** and **Azure** to automate testing and deployment for a 5-person engineering team handling **10+ deployments per week**, resolving stale deployment issues that were blocking new releases due to Azure slot limits and **reducing deployment time by 50%**.

University of Oregon

Sep 2021 - Jun 2024

Graduate Teaching Assistant

Eugene, OR

- Applied innovative teaching techniques including **live coding demos** and **one-on-one mentoring** during office hours across **Web Development I & II**, **Computer Networks**, and **Database Systems**, boosting class participation by **60%**, verified through end-of-semester participation metrics and student feedback, strengthening undergraduate students' foundations in industry-relevant technical skills.
- Created a Jest-based testing script, **reducing grading time by 80%** and administrative **overhead by 60%**, significantly improving course management efficiency.

Advanta Seeds

Jun 2023 - Sep 2023

Global Software Developer Intern

College Station, TX

- **Reduced system migration time by 50%** through modular code refactoring, ensuring timely project delivery ahead of schedule.
- **Enhanced data accuracy and application performance** by leading an **SQL database** refactoring initiative, **reducing data redundancy by 40%**, verified through data integrity checks and query performance benchmarking before and after implementation.

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DePeuter Packaging Solutions

Apr 2019 - Jan 2020

Mechanical Engineer

Houston, TX

- **Enhanced throughput by 40%** through PLC and robot programming, measured via production line output tracking before and after implementation; developed HMIs that visualized machine states and production workflows.
- Designed end-of-arm tools and structural components using **AutoCAD, SolidWorks, and Fusion 360**, achieving at least **20% reduction in production costs**, verified through materials and cycle time analysis.

PROJECTS

Pokemon-fly — Connectome-Driven Pokémon Player

Sep 2026

Python, FastAPI, WebSockets, NumPy, SciPy, TypeScript, Next.js, React, Three.js, Docker

- Built a real-time neural simulation of a fruit fly's brain and nerve cord to autonomously play Pokémon Red, modeling **39,991 real neurons and 1.7M synapses** from two EM-reconstructed *Drosophila* connectomes, by implementing a sparse rate-based recurrent network in NumPy/SciPy with dopamine-gated Hebbian plasticity at the mushroom body's learning synapses.
- Engineered a real-time backend streaming synchronized game video, audio, and live 3D neural activity to a web dashboard, achieving audio/video sync within a **150ms lookahead window** across four concurrent loops, by coordinating an emulator, decision engine, and stream pipeline with asyncio and thread-safe locking in FastAPI over WebSockets.
- Designed an exploit-resistant reinforcement signal translating raw Game Boy memory reads into a dopamine-driven reward system, preventing reward-farming from revisits or backtracking through permanent-record novelty tracking across **5 independent signal types** (tile, map, scene, text, story-route progress), deployed via a Dockerized FastAPI/Next.js/Three.js stack.

Protos — Multi-Cloud Monitoring Dashboard

Jun 2026 — Jul 2026

Java, Spring Boot, Kafka, PostgreSQL, TypeScript, Next.js, React, Node.js, Electron, Docker, GitHub Actions

- Built a multi-cloud monitoring platform ingesting live resource status from GitHub, Azure, and AWS, enabling the dashboard and alerting to scale independently as new cloud providers are added, by architecting an event-driven backend of **8 Spring Boot microservices** that publish resource-state changes to Kafka, consumed separately by an aggregator (dashboard + WebSocket push) and a notifier (alerting) service.
- Secured a multi-tenant SaaS backend against credential leakage and stale-session access, verified by re-checking account status against the database on every request rather than only at login, by implementing a JWT-authenticated Spring Cloud Gateway with per-user tenant isolation and **AES-GCM encryption** of stored cloud credentials (Azure/AWS keys, GitHub tokens).
- Automated AI-generated repository documentation for arbitrary codebases, validated by **139 JUnit tests across 6 backend services** covering auth, rate limiting, and OAuth flows, by building an async pipeline that batches source files by character budget, calls the Claude API per batch, and synthesizes the results into a single Markdown doc with commit-hash-based caching.

ETO — Social & Business Review Platform

May 2026 — Jun 2026

Java, Spring Boot, TypeScript, Next.js, PostgreSQL, Redis, Docker, Kubernetes

- Architected a **9-service Spring Boot microservices platform** with a Spring Cloud Gateway handling JWT authentication, IP-based rate limiting, and WebSocket proxying for real-time notifications and live chat.
- Designed cross-service data flows including a **privacy-enforced social graph**, geospatial business search, per-ethnicity review analytics, and an **append-only coin ledger** supporting a full campaign and redemption reward system.
- Containerized the full stack with Docker Compose across 9 services, PostgreSQL, Redis, and MinIO object storage with presigned S3 upload flows; authored **Kubernetes manifests** for cloud deployment.

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Tut — AI Learning Assistant

Jan 2026

C#, Python, JavaScript, TypeScript, ASP.NET, FastAPI, Docker, Ollama

- Architected a real-time full-stack system that streams coding activity over WebSockets and provides **live feedback based on behavioral signals**.
- Designed an event-driven messaging protocol (session.start, code.snapshot, hint) with **versioned envelopes** for extensibility.
- Integrated **streaming LLM (Ollama) responses** to deliver incremental feedback in real-time rather than batch responses.

EDUCATION

Master of Science - Computer Science

Sep 2021 - Jun 2024

University of Oregon - Eugene, OR - GPA: 3.68

Bachelor of Science - Manufacturing and Mechanical Engineering

Sep 2015 - Dec 2018

Texas A&M University - College Station, TX - GPA: 3.53